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Reducing False-Positive Rates in Machine Learning-Based Transaction Fraud Detection

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Executive Summary

This report examines the problem of high false-positive rates in the machine learning models that banks use to detect transaction fraud, in the context of fraud analytics at Scotiabank. Fraudulent transactions typically represent well under 1% of all card activity, and this extreme class imbalance causes conventional classifiers to generate large volumes of alerts on legitimate transactions. Industry research indicates that more than 90% of transaction monitoring alerts at most banks are false positives (McKinsey & Company, 2022), which consumes analyst capacity, and falsely declined transactions have been estimated to cost card issuers far more than the fraud they prevent (Javelin Strategy & Research, 2015).

The scope of this report is a secondary research analysis. It reviews the academic literature on learning from imbalanced data and compares three techniques: data-level approaches that resample the training data, such as the Synthetic Minority Over-sampling Technique (SMOTE); algorithm-level approaches, such as cost-sensitive learning and ensemble methods; and decision-level approaches, such as cost-based threshold tuning. The analysis also evaluates which performance metrics are appropriate for imbalanced problems, since the literature shows that accuracy and even Receiver Operating Characteristic (ROC)-based measures can make a poor fraud model look strong.

The analysis concludes that no single technique solves the false-positive problem on its own, and that the choice of evaluation metric is just as important as the choice of modelling technique. The report recommends that the fraud analytics team adopt precision-recall and alert-based metrics in model validation, pilot cost-sensitive training and cost-based threshold

optimization on existing models first, and treat resampling methods such as SMOTE as experimental tools that require careful cross-validation practices.

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1. Introduction

The Bank of Nova Scotia (Scotiabank) is one of Canada's largest banks, with roughly 25 million customers across the Americas in personal and commercial banking, wealth management, and capital markets. Its data and analytics teams build machine learning models used throughout the bank for credit decisions, customer engagement, and fraud detection. Within fraud analytics, the Data Science Intern role consists of exploratory data analysis, feature engineering, training and validation of models, and documentation of the decisions behind them.

Fraud detection in retail banking is a difficult machine learning problem because models must flag fraudulent transactions in near real time without blocking legitimate ones. The central technical issue is class imbalance. Fraudulent transactions usually make up well under 1% of all transactions in payment card datasets, so conventional classifiers end up optimizing for the majority class and producing alerts with very high false-positive rates (Dal Pozzolo et al., 2018). This creates real operational problems. Fraud analysts have to manually review every alert, and when most alerts turn out to be legitimate activity, their time is consumed by low-value work. Meanwhile, customers whose real transactions are declined become frustrated, which erodes their trust in the bank. There is a substantial academic literature on how to handle imbalanced classification, including resampling methods such as SMOTE, cost-sensitive learning, and ensemble methods built for imbalanced data (Krawczyk, 2016).

This report analyzes the trade-offs among these techniques in the context of transaction fraud detection, focusing on the precision-recall trade-off and on how the choice of evaluation metric affects the way model performance is interpreted. The objectives of the report are: first, to explain why standard classification approaches produce high false-positive rates on fraud data;

second, to compare data-level, algorithm-level, and decision-level techniques for reducing false positives against criteria that matter in a banking environment; and third, to recommend an approach that could lower the false-positive rate of Scotiabank's fraud models without losing too much accuracy on actual fraud cases. If successful, this would reduce analyst review volume and reduce the number of legitimate transactions that are declined. The report first reviews the relevant literature, then describes the method of analysis, presents the analysis itself, and closes with conclusions and recommendations for the fraud analytics team.

2. Literature Review / Background Information

The Class Imbalance Problem

Class imbalance refers to classification problems where one class heavily outnumbers the other. He and Garcia (2009) survey the field and show that standard learning algorithms implicitly assume balanced classes and equal misclassification costs; when those assumptions fail, classifiers become biased toward the majority class. In the most extreme case, a model that simply labels every transaction as legitimate achieves near-perfect accuracy while detecting no fraud at all. Krawczyk (2016) notes that imbalance is rarely the only difficulty: minority classes also tend to be heterogeneous, overlap with the majority class, and evolve over time, all of which apply to fraud.

Transaction fraud is one of the most extreme practical examples of imbalance. In the widely used public dataset released by the Université Libre de Bruxelles (ULB) Machine Learning Group, only 492 of 284,807 card transactions (0.172%) are fraudulent (Dal Pozzolo et al., 2015). Dal Pozzolo et al. (2018) add two further complications from production fraud systems: verification latency, since the true label of a transaction is often only confirmed days or weeks later when an investigator or cardholder reports the outcome, and concept drift, since both spending behaviour and fraud strategies change over time. A technique that reduces false positives must therefore work under imbalance, delayed labels, and drift simultaneously.

Data-Level Approaches: Resampling

One way to address class imbalance is to rebalance the training data before a model is fit. Chawla et al. (2002) introduced the Synthetic Minority Over-sampling Technique (SMOTE), which creates synthetic examples of the minority class by interpolating between neighbouring

minority instances instead of duplicating existing ones. They showed that combining this over-sampling of the minority class with under-sampling of the majority class led to stronger classifier performance than under-sampling alone. Fernández et al. (2018), reviewing fifteen years of work on SMOTE, conclude that it remains a standard baseline but caution that synthetic examples can amplify noise and blur class boundaries, and that resampling must be applied only to training folds during cross-validation to avoid leaking information into the evaluation.

Under-sampling the majority class is computationally favoured because it shrinks the training set, but it discards information and changes the class before the model learns. Dal Pozzolo et al. (2015) show that models trained on under-sampled data produce distorted probability estimates and propose a calibration correction for this bias, which matters when alert thresholds are set on predicted probabilities. In their practitioner-oriented study of a real fraud detection system, Dal Pozzolo et al. (2014) found resampling useful but warned that it must be paired with evaluation metrics suited to imbalanced and drifting data.

Algorithm-Level Approaches: Cost-Sensitive Learning and Ensembles

Instead of changing the data, cost-sensitive methods change the objective the model optimizes. Elkan (2001) provides the theoretical foundation, showing that once the costs of false positives and false negatives are specified, the optimal decision threshold follows directly from the cost matrix, and that training with class weights is formally equivalent to certain forms of resampling. In fraud detection the costs are asymmetric but not one-sided: a missed fraud costs the bank the fraud amount, while a false positive costs analyst review time and customer goodwill. Bahnsen et al. (2016) demonstrate on real card data that example-dependent cost-sensitive models, which weight each transaction by its monetary amount, change which model is preferred compared with cost-blind evaluation.

Ensemble methods are the other major algorithm-level technique. Krawczyk (2016) and West and Bhattacharya (2016) both report that ensembles, particularly random forests and bagging variants combined with sampling, are among the most consistently strong performers on imbalanced financial data. West and Bhattacharya (2016), in a review of intelligent fraud detection across 49 studies, also note that comparing published results is difficult because studies use different datasets, imbalance ratios, and metrics, which is a key reason this report treats metric selection as part of the problem. More recent work has explored sequence models: Jurgovsky et al. (2018) frame fraud detection as sequence classification with recurrent neural networks and report gains on face-to-face transaction sequences, though at a cost in complexity and interpretability that matters for model risk review in banks.

Evaluation Metrics for Imbalanced Problems

A large share of the false-positive problem comes from measuring the wrong thing. Dal Pozzolo et al. (2014) studied credit card fraud detection using real transaction data and identified three main challenges: too few fraud cases to learn from, fraud patterns that keep changing, and the difficulty of measuring how well a model works. They argued that conventional accuracy-based metrics are misleading in this setting, because a model can appear highly accurate while still flagging large volumes of legitimate activity, and they recommended alert-based measures, such as precision within the top-ranked alerts that analysts can actually review, because these reflect how investigation teams work in practice.

Saito and Rehmsmeier (2015) make the complementary statistical argument: on imbalanced data, the precision-recall curve is more informative than the ROC curve, because the ROC curve's false-positive rate is computed over an enormous negative class and can look excellent even when alerts are dominated by false positives. The practical implication is that area

under the precision-recall curve (AUPRC) and precision at a fixed alert budget are better aligned with the operational cost of false positives than accuracy or ROC-AUC.

Operational Impact of False Positives

McKinsey & Company (2022) report that for most banks, more than 90% of transaction monitoring alerts turn out to be false positives, so the bulk of investigator capacity is spent clearing legitimate activity. On the customer side, Javelin Strategy & Research (2015) estimated that United States card issuers declined legitimate transactions worth $118 billion in a year, that false declines affect far more cardholders than actual fraud does, and that nearly four in ten falsely declined cardholders stopped using the card afterward. These findings frame false positives not as a cosmetic modelling issue but as a direct driver of operating cost and customer attrition, which is the perspective taken in the analysis below.

3. Materials and Methods

This report is based on secondary research; no primary experiments have been conducted. Sources were selected in three groups. First, foundational and survey papers on imbalanced classification (Chawla et al., 2002; He & Garcia, 2009; Krawczyk, 2016; Fernández et al., 2018) establish the available techniques. Second, fraud-specific studies on real or realistic transaction data (Dal Pozzolo et al., 2014, 2015, 2018; Bahnsen et al., 2016; Jurgovsky et al., 2018; West & Bhattacharya, 2016) provide evidence about what works under fraud conditions. Third, industry research (Javelin Strategy & Research, 2015; McKinsey & Company, 2022) quantifies the operational cost of false positives. Peer-reviewed sources were preferred, and fraud-specific evidence was weighted more heavily than results on generic benchmark datasets.

The analysis proceeds in four steps that another analyst could repeat. Step one states the problem and defines the criteria a solution must meet. Step two uses a worked numerical example to show why accuracy-based evaluation hides the false-positive problem. Step three compares the candidate techniques, organized as data-level, algorithm-level, and decision-level interventions, against the criteria from step one, using the findings reported in the reviewed literature. Step four assesses which evaluation metrics should be used to judge any of these techniques.

4. Analysis / Results and Analysis

Problem Statement and Criteria

The problem analyzed in this report is that machine learning models used for transaction fraud monitoring generate alert sets dominated by false positives, wasting analyst capacity and harming legitimate customers, because the models are trained and evaluated in ways that ignore extreme class imbalance and asymmetric error costs. A solution is considered successful if it meets four criteria. First, it must materially increase alert precision, meaning a higher share of flagged transactions are fraudulent. Second, it must do so without a material loss of recall, since missed fraud is a direct financial loss. Third, it must be feasible to implement within a bank's constraints, including model risk review, explainability expectations, and existing model pipelines. Fourth, its evaluation must use metrics that remain meaningful under imbalance, since otherwise improvements cannot be demonstrated credibly.

Why Accuracy Hides the Problem

A worked example shows why the most familiar metric fails. Consider one million transactions with a fraud prevalence of 0.15%, a simplified figure in line with public fraud datasets (Dal Pozzolo et al., 2015): 1,500 transactions are fraudulent and 998,500 are legitimate. Model A is a degenerate classifier that flags nothing. Model B is a realistic classifier that catches 80%of fraud (1,200 cases) with a false-positive rate of 0.5% (4,993 false alerts). Table 1 compares the two.

Measure	**Model A: flag nothing**	**Model B: realistic classifier**
Frauds caught (true positives)	0	1,200
Frauds missed (false negatives)	1,500	300

False alerts (false positives)	0	4,993
Accuracy	99.85%	99.47%
Recall (fraud caught)	0%	80%
Precision (alerts that are fraud)	undefined (no alerts)	19.4%

Table 1. Performance of two hypothetical fraud classifiers on one million transactions with 0.15% fraud prevalence. Model A flags no transactions; Model B catches 80% of fraud with a 0.5% false-positive rate. Model A wins on accuracy while providing no fraud protection.

Model A achieves higher accuracy, 99.85%, while detecting zero fraud. Model B, which provides real protection, scores lower on accuracy, and even so, four out of every five of its alerts are false. This is the same pattern reported at industry scale, where more than 90% of transaction monitoring alerts are false positives (McKinsey & Company, 2022). Two conclusions follow. Accuracy must be excluded from model selection for this problem, and even a model with a false-positive rate that sounds tiny in relative terms produces an absolute alert volume that overwhelms review capacity. The relevant management variable is precision at the alert volume analysts can actually handle (Dal Pozzolo et al., 2014).

From Precision to Analyst Workload

Translating that management variable into staffing terms shows what is at stake for the fraud analytics team. Suppose that the team's monitoring models show roughly 10,000 alerts on a typical day, and that a fraud analyst can carefully clear 100 alerts per shift. Reviewing every alert would then require about 100 analyst-shifts per day, which is far beyond any realistic team size. In practice, the team would investigate only the highest-ranked alerts, which for simplicity will be the top 1,000. The question is therefore not how the model scores across all transactions, but how many of those top 1,000 reviewed alerts are genuinely fraudulent. At the 19.4% precision of Model B, 194 of the reviewed alerts would be fraud and 806 would be wasted reviews. Raising

precision at the same alert budget to 40% would lift the number of true frauds reviewed to about 400 and cut wasted reviews to 600, without adding a single analyst or changing how many alerts the team reads. This is the practical pay-off the recommended metrics are designed to capture: model quality is measured in frauds confirmed per analyst-shift, which is the unit the team is staffed around. The specific amounts here are fictional and would be replaced with Scotiabank's own daily alert counts and per-analyst review capacity, but the relationship holds for whatever values the team substitutes.

Comparison of Techniques for Reducing False Positives

Table 2 summarizes the candidate techniques from the literature review, organized by where they intervene: in the data, in the learning algorithm, or in the decision rule applied to model scores.

Technique	Mechanism	Strengths	Limitations
Random under-sampling	Discards majority-class examples until classes are balanced	Simple; fast training; strong baseline	Throws away data; distorts predicted probabilities unless calibrated (Dal Pozzolo et al., 2015)
SMOTE over-sampling	Creates synthetic minority examples by interpolation (Chawla et al., 2002)	No data discarded; improves minority sensitivity	Can amplify noise and blur class boundaries; must be applied inside training folds only (Fernández et al., 2018)
Cost-sensitive learning	Weights errors by misclassification cost during training (Elkan, 2001)	No synthetic data; aligns model with business costs; works with existing pipelines	Requires credible cost estimates; costs vary by transaction (Bahnsen et al., 2016)
Ensemble methods	Combines many classifiers, often with per-model sampling	Consistently strong on imbalanced financial data (Krawczyk, 2016; West & Bhattacharya, 2016)	More complex to explain in model risk review; higher compute

Threshold tuning	Sets the alert threshold from the cost matrix or alert capacity rather than 0.5	No retraining needed; directly controls alert volume	Cannot fix a poorly ranked model; needs calibrated probabilities

Table 2. Comparison of techniques for reducing false positives in imbalanced classification, organized by intervention level: data-level (rows 1-2), algorithm-level (rows 3-4), and decision-level (row 5). Citations refer to the supporting literature reviewed in this report.

Evaluated against the four criteria, the techniques differ more in operational fit than in raw capability. Resampling methods score well on simplicity and on published performance: Chawla et al. (2002) showed that SMOTE combined with under-sampling outperforms under-sampling alone, and the goal of resampling in fraud detection is to make a model sensitive enough to catch the same amount of fraud while flagging fewer legitimate transactions. However, resampling carries two governance risks in a banking context. Synthetic transactions are artifacts of an interpolation procedure rather than observed behaviour, which complicates model documentation, and incorrectly applied resampling, such as resampling before cross-validation splits, inflates reported performance (Fernández et al., 2018). Under-sampling additionally biases predicted probabilities, which corrupts any threshold set on those probabilities unless the calibration correction of Dal Pozzolo et al. (2015) is applied.

Cost-sensitive learning fits the banking constraints more naturally. It requires no synthetic data, is supported as a class-weight option in standard libraries, and ties the model objective to quantities the business already tracks: the cost of an analyst review and the expected loss of a missed fraud. Elkan (2001) shows the formal equivalence between cost weighting and resampling, which means cost-sensitive training can deliver similar statistical benefits with a cleaner audit trail. Bahnsen et al. (2016) extend this by weighting each transaction by its own

amount, which matches the intuition that missing a $5,000 fraud is far worse than missing a $50 one.

Putting concrete figures on these costs clarifies the asymmetry the model should encode. The cost of a false positive is the sum of an analyst's review time and the harder-to-quantify cost of a wrongly declined sale, including lost interchange revenue and the risk of customer attrition. Javelin Strategy & Research (2015) found that nearly four in ten falsely declined cardholders stopped using the card, so even a conservative estimate of the value lost on a fraction of those customers can exceed the labour cost of the review itself. The cost of a false negative is the fraud the bank absorbs, which Bahnsen et al. (2016) argue should be the transaction's own amount rather than a flat average. A starting point for a pilot might value a false positive at roughly $4 of review labour plus an expected $15 to $25 of declined sale and attrition cost, against a false negative valued at the transaction amount; even these rough values push the optimal alert threshold far below the default of 0.5. Reaching agreement on the cost estimates is the main practical difficulty, since it requires input from fraud operations rather than data science alone.

Decision-level threshold tuning is the cheapest intervention because it requires no retraining at all. Most classifiers default to a 0.5 probability threshold, which implicitly assumes equal misclassification costs; under realistic fraud costs the optimal threshold is far lower, and conversely, where analyst capacity is the limiting factor, the threshold can be set to deliver exactly the alert volume the team can review (Elkan, 2001). Appendix A works through a numerical example using the cost figures above. The limitation is that threshold tuning only re-cuts the ranking a model already produces; if the model ranks fraud poorly, no threshold rescues it. Ensembles and sequence models (Jurgovsky et al., 2018) improve the ranking itself but raise

explainability and maintenance costs, which makes them a secondary option rather than a starting point.

Choosing the Right Evaluation Metrics

None of the techniques above can be adopted responsibly unless the metrics used to judge them are valid under imbalance. Table 3 assesses the common options.

Metric	What it measures	Suitability for fraud detection
Accuracy	Share of all transactions classified correctly	Misleading: dominated by the majority class (Dal Pozzolo et al., 2014)
ROC-AUC	Ranking quality across all thresholds, using the false-positive rate	Overly optimistic under heavy imbalance (Saito & Rehmsmeier, 2015)
Precision	Share of alerts that are actually fraud	Directly measures analyst workload quality; threshold-dependent
Recall	Share of fraud that is caught	Measures protection; must be reported alongside precision
AUPRC	Area under the precision-recall curve	Recommended summary metric for imbalanced data (Saito & Rehmsmeier, 2015)
Precision at alert budget	Precision within the top-k alerts analysts can review	Closest to operational reality (Dal Pozzolo et al., 2014)

Table 3. Evaluation metrics and their suitability for transaction fraud detection. Threshold-free metrics (ROC-AUC, AUPRC) summarize ranking quality; precision, recall, and precision at an alert budget describe performance at an operating point.

The combination recommended by the literature is AUPRC as the threshold-free summary, reported together with precision and recall at the operating threshold, plus precision within the top-ranked alerts corresponding to the analyst team's daily review capacity (Dal Pozzolo et al., 2014; Saito & Rehmsmeier, 2015). This last metric deserves emphasis because it converts model quality into a unit that fraud operations manages. It also makes the cost of false positives visible to model developers in a way ROC-AUC never does.

Dataset Characteristics and Their Consequences

Figure 1 shows the class distribution of the public ULB credit card fraud dataset, in which 492 of 284,807 transactions are fraudulent (Dal Pozzolo et al., 2015). The bar chart requires a logarithmic scale simply to make the fraud class visible, which is a useful way to communicate to non-technical stakeholders why fraud modelling cannot be treated as a standard classification task.

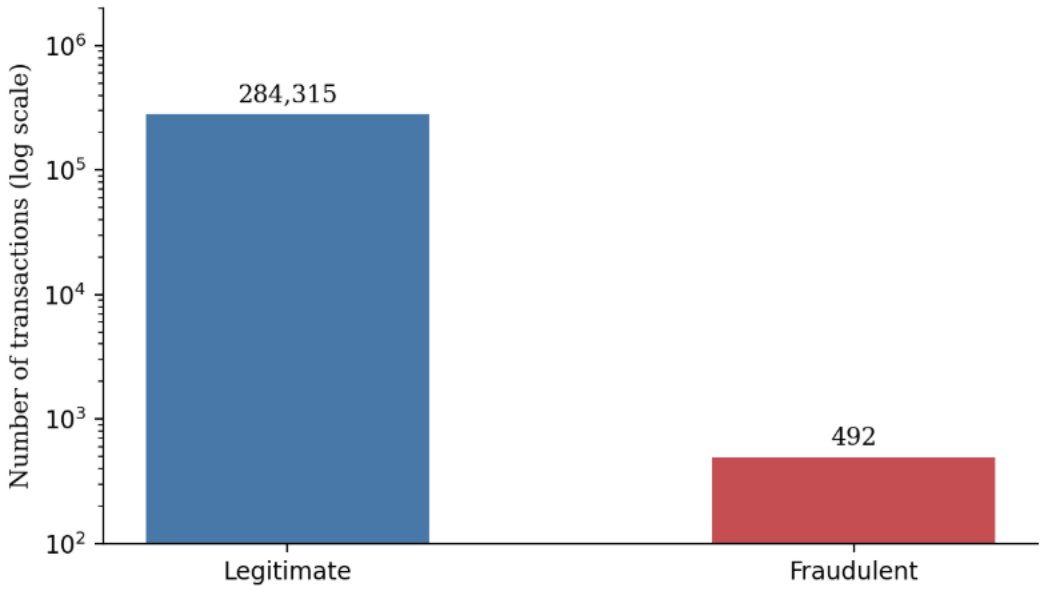


Figure 1. Class distribution of the ULB credit card fraud dataset: 284,315 legitimate versus 492 fraudulent transactions (0.172% fraud). The vertical axis uses a logarithmic scale; on a linear scale the fraud class would not be visible. Data from Dal Pozzolo et al. (2015).

Two further properties of production fraud data condition the analysis. Verification latency means recent transactions have unreliable labels, so retraining sets must be constructed from confirmed outcomes, and concept drift means any tuned threshold or resampling ratio degrades over time (Dal Pozzolo et al., 2018). This favours techniques that are cheap to re-run, such as threshold re-optimization and class-weight retraining, over techniques that require

regenerating synthetic datasets, and it implies that false-positive performance must be monitored continuously rather than validated once at deployment.

5. Discussion

The central finding of the analysis is that high false-positive rates in fraud detection are not primarily a model quality problem but a problem of misaligned objectives and metrics. The worked example in Table 1 shows that a bank optimizing accuracy would prefer a useless model to a useful one, and the industry evidence that over 90% of alerts are false positives (McKinsey & Company, 2022) suggests that many production systems are still operated far from their cost-optimal threshold. For the fraud analytics team, this is an encouraging finding: meaningful reductions in analyst workload may be available through evaluation and thresholding changes that do not require new modelling techniques at all.

The comparison of techniques also suggests an ordering of interventions that fits how banks govern models. Threshold tuning and cost-sensitive retraining preserve the existing model architecture and audit trail, so they can move through model risk review quickly; resampling experiments and ensemble or sequence models offer further gains but carry documentation, validation, and explainability burdens (Fernández et al., 2018; Jurgovsky et al., 2018). This ordering matters for a practical reason identified by West and Bhattacharya (2016): published studies disagree about which technique is best partly because they evaluate on different data and metrics, so a bank should expect to run its own comparisons, and it should start with the interventions that are cheapest to trial.

These results must be interpreted with limitations in mind. The quantitative figures cited come from public datasets and from other institutions; Scotiabank's transaction mix, fraud prevalence, and cost structure will differ, so the specific numbers in Table 1 and Appendix A are illustrative rather than predictive. The industry statistics from Javelin Strategy & Research (2015) and McKinsey & Company (2022) describe the United States market and anti-money-

laundering monitoring respectively, so they bound the problem rather than measure it for one bank.

6. Conclusions and Recommendations

Three conclusions follow from the analysis. First, extreme class imbalance makes accuracy and ROC-based evaluation actively misleading for fraud detection, and metric choice is therefore a root cause of, not just a symptom of, high false-positive rates. Second, the main techniques: resampling, cost-sensitive learning, ensembles, and threshold tuning; all have credible published evidence behind them. However, they differ sharply in how well they fit banking constraints such as model risk review, probability calibration, and drift; decision-level and cost-based interventions fit best. Third, because fraud data drifts and labels arrive late, false-positive control is an ongoing operating practice rather than a one-time modelling fix.

Based on these conclusions, this report makes three recommendations to the fraud analytics team. First, adopt AUPRC, precision and recall at the operating threshold, and precision within the daily alert budget as the standard evaluation metrics for fraud models, replacing accuracy and standalone ROC-AUC in validation reports (Dal Pozzolo et al., 2014; Saito & Rehmsmeier, 2015). Second, pilot cost-based threshold optimization on the existing production models, using review-cost and loss estimates agreed with fraud operations, since this requires no retraining and can reduce alert volume immediately (Elkan, 2001). Third, trial cost-sensitive retraining, including amount-weighted variants (Bahnsen et al., 2016), before introducing resampling pipelines; where SMOTE or under-sampling is used in experiments, enforce resampling inside training folds only and apply probability calibration (Dal Pozzolo et al., 2015; Fernández et al., 2018).

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Appendix A

This appendix illustrates the threshold tuning analysis referenced in the Analysis section, following Elkan (2001). Suppose the cost of a false positive (one analyst review plus customer friction) is estimated at $20, and the expected cost of a false negative (an average missed fraud) is $500. For a classifier that outputs a calibrated probability p that a transaction is fraudulent, the expected cost of not alerting is 500p, and the expected cost of alerting is 20(1 − p). Alerting is worthwhile when 500p exceeds 20(1 − p), which simplifies to p greater than 20 / (20 + 500), approximately 0.038.

In other words, under these costs the bank should alert whenever the model assigns even a 3.8% fraud probability, far below the 0.5 default threshold most classifiers apply. The default threshold implicitly assumes false positives and false negatives cost the same, which is wrong by more than an order of magnitude. The result is sensitive to the cost estimates: if the review cost rises to $50, the threshold rises to 50 / 550, approximately 0.091, and if analyst capacity is fixed, the threshold can instead be chosen as the score of the k-th highest-ranked transaction, where k is the number of alerts the team can review per day. This calculation requires calibrated probabilities, which is why probability distortion from under-sampling (Dal Pozzolo et al., 2015) matters in practice. These dollar values are illustrative; actual cost estimates would be agreed with fraud operations before deployment.